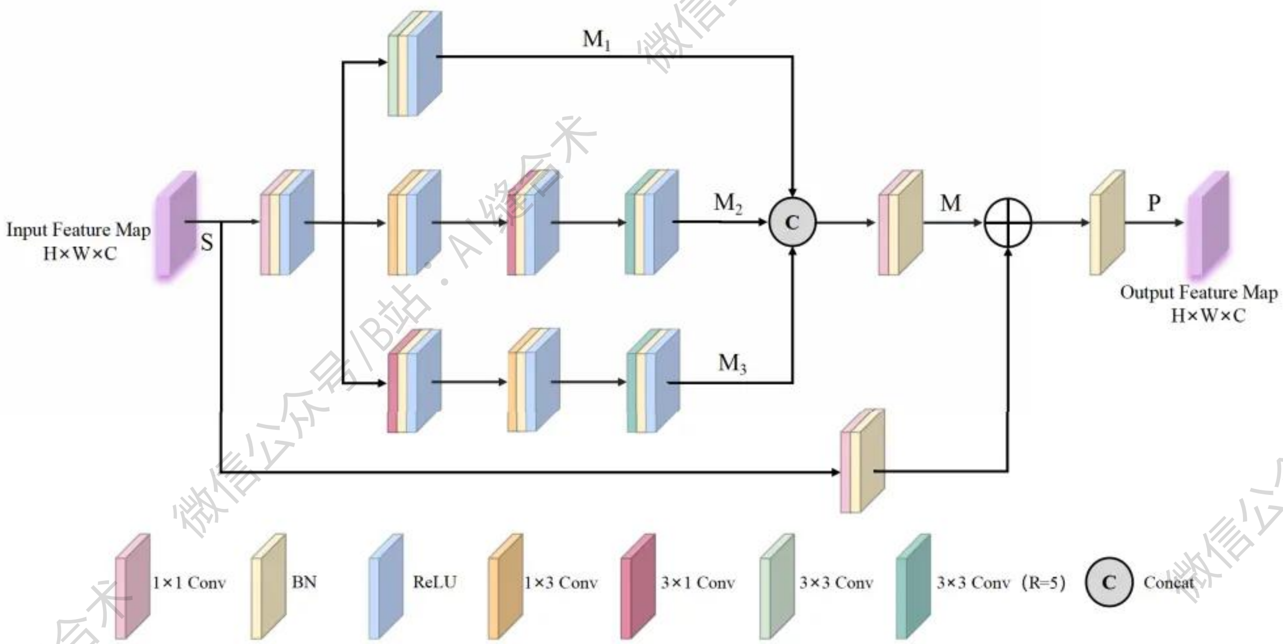
核心速览



论文题目： Lightweight Multiscale Feature Fusion and Multireceptive Field Feature Enhancement for Small Object Detection in the Aerial Images

中文题目： 航空图像小目标检测的轻量级多尺度特征融合与多感受野特征增强

论文链接： https://ieeexplore.ieee.org/abstract/document/11087713

所属单位： 南京邮电大学等

核心速览： 本文提出了一种轻量级多尺度特征融合与多感受野特征增强算法（LMFFMFFE），用于解决无人机平台在复杂环境下小目标检测中模型参数与检测精度难以平衡的问题。

论文内容详细解读：https://mp.weixin.qq.com/s/-SZGfeooLJCMin0tpwny3w?token=644106154&lang=zh_CN

```
MRFFE(
  (conv1x1_in): Conv2d(32, 8, kernel_size=(1, 1), stride=(1, 1), bias=False)
  (bn_in): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (relu_in): ReLU(inplace=True)
  (m1_conv1): Conv2d(8, 8, kernel_size=(1, 3), stride=(1, 1), padding=(0, 1), bias=False)
  (m1_bn1): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m1_relu1): ReLU(inplace=True)
  (m1_conv2): Conv2d(8, 8, kernel_size=(3, 1), stride=(1, 1), padding=(1, 0), bias=False)
  (m1_bn2): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m1_relu2): ReLU(inplace=True)
  (m1_conv3): Conv2d(8, 8, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
  (m1_bn3): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m1_relu3): ReLU(inplace=True)
  (m2_conv1): Conv2d(8, 8, kernel_size=(3, 1), stride=(1, 1), padding=(1, 0), bias=False)
  (m2_bn1): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m2_relu1): ReLU(inplace=True)
  (m2_conv2): Conv2d(8, 8, kernel_size=(1, 3), stride=(1, 1), padding=(0, 1), bias=False)
  (m2_bn2): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m2_relu2): ReLU(inplace=True)
  (m2_conv3): Conv2d(8, 8, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
  (m2_bn3): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m2_relu3): ReLU(inplace=True)
  (m3_conv1): Conv2d(8, 8, kernel_size=(3, 3), stride=(1, 1), padding=(2, 2), dilation=(2, 2), bias=False)
  (m3_bn1): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m3_relu1): ReLU(inplace=True)
  (m3_conv2): Conv2d(8, 8, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
  (m3_bn2): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m3_relu2): ReLU(inplace=True)
  (m3_conv3): Conv2d(8, 8, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1), bias=False)
  (m3_bn3): BatchNorm2d(8, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (m3_relu3): ReLU(inplace=True)
  (conv1x1_out): Conv2d(24, 32, kernel_size=(1, 1), stride=(1, 1), bias=False)
  (bn_out): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True, track_running_stats=True)
  (relu_out): ReLU(inplace=True)
  (shortcut): Identity()
  (bn_shortcut): Identity()
)
Input shape: torch.Size([1, 32, 256, 256])
Output shape: torch.Size([1, 32, 256, 256])
```